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Complex-valued proximal neural network method for solving complex-valued mixed variational inequalities

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ABSTRACT

This paper proposes a complex-valued proximal neural network (CPNN) method for solving complex-valued mixed variational inequalities (CMVIs) directly in the complex-domain. Under standard convexity assumptions, we establish the nonexpansiveness of the complex-valued proximal mapping. Based on this proximal formulation, we further establish the existence and uniqueness of solutions to the CMVIs and derive a corresponding error bound under the assumptions of $\theta - \Phi$ strong pseudomonotonicity and Lipschitz continuity. We then prove that the equilibrium points of the proposed CPNN coincide with the solutions of the CMVIs, and that the trajectories of the network are well posed. By combining the complex chain rule with Lyapunov analysis, we establish global exponential stability of the continuous-time dynamics and prove linear convergence of its discrete-time implementation under suitable discretization conditions. Finally, numerical experiments demonstrate the effectiveness of the proposed method.

1. Introduction

In many engineering and scientific applications, variables are often represented by complex numbers, including probability analysis, signal processing, and medical imaging (Bi et al., 2025; Lustig et al., 2007; Vakhania & Kandelaki, 1997; Zhang & Ma, 1997). In these cases, both magnitude and phase carry important information, and directly modeling the unknowns as complex-valued vectors can better preserve intrinsic structures that might be concealed by real-valued isomorphisms. For instance, complex-domain modeling has been shown to be advantageous in wind speed profile prediction (Goh et al., 2006).

Meanwhile, neurodynamic networks have been widely studied as continuous-time solver architectures for optimization and equilibrium problems due to their inherent parallelism and real-time feasibility (Hu & Wang, 2006; Vuong & Strodiot, 2020). As a unified modeling tool for constrained optimization and equilibrium systems with non-smooth regularization, mixed variational inequalities (MVIs) have attracted increasing attention. In the real-domain, a growing body of work has established convergence and stability properties of neurodynamic solvers for MVIs, including global exponential stability and linear convergence under strong pseudomonotonicity conditions (Ju et al., 2022, 2021). More recently, refined convergence-time guarantees have been pursued, such as fixed-time neural dynamics and predefined-time designs for MVIs (Wen et al., 2024; Zheng et al., 2024). In contrast, for complex-valued mixed variational inequalities (CMVIs),

especially those involving composite nonsmooth terms and complex-valued proximal mapping, systematic results on existence and uniqueness, residual-based error bounds, global exponential stability, and linear convergence remain comparatively scarce. This gap motivates us to study CMVIs and to develop solver architectures that operate intrinsically in the complex-domain with explicit stability and convergence guarantees.

A common way to handle complex-variable problems is to rewrite $z = x + iy$ and reformulate the problem in $\mathbb{R}^{2n}$ (Qin et al., 2016). This reformulation is mathematically equivalent and does not by itself discard information. However, for nonsmooth regularizers, it changes the structural form of the model. For example, the complex ℓ_1 regularizer $\Phi(z) = \sum_{j=1}^n |z_j|$ becomes $\sum_{j=1}^n \sqrt{x_j^2 + y_j^2}$ in $\mathbb{R}^{2n}$, which is a group-type penalty acting on the two-dimensional blocks (x_j, y_j) . Consequently, the associated proximal step is no longer separable with respect to the real and imaginary parts individually. This motivates us to develop proximal neurodynamic networks directly in the complex-domain, where the model and its proximal operator can be expressed in their native form. For this purpose, Wirtinger calculus, complex subdifferential, and the complex chain rule provide a convenient analytical framework (Koor, Qiu, Kwek, & Rebentrost, 2023; Liu, Jiang, Zhang, & Mei, 2020).

Related to complex-valued modeling, complex-valued neural networks (CVNNs), as extensions of real-valued neural networks, have attracted sustained interest; systematic overviews of architectures, activation functions, learning mechanisms, and applications can be found in

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Hirose (2006), Lee et al. (2022), Liu (2025), and representative advances on modeling components are reported in Scardapane et al. (2018). From the viewpoint of dynamical properties, the existence, uniqueness, and asymptotic stability of equilibrium points in CVNNs were analyzed in Li and Huang (2018). Moreover, complex-domain neurodynamic approaches have been developed to solve complex-variable constrained optimization, complex-variable variational inequalities, and complex-variable programming problems, including projection dynamical models equipped with stability analysis (Lee, Hasegawa, & Gao, 2022; Liu, 2025; Sorber, Barel, & Lathauwer, 2012; Zhang, Xia, & Wang, 2015a; Zheng, Ju, Zhang, & Xu, 2025). However, these advances have primarily been confined to smooth objective functions, leaving aside the composite nonsmooth regularization terms that are crucial for applications such as sparse recovery. In recent years, deep learning and neural-network-based regularization have also become prominent tools for imaging inverse problems (Habring & Holler, 2024; Mukherjee et al., 2023). However, such data-driven methods usually require large datasets and often lack rigorous theoretical guarantees. In this work, our focus is on model-driven complex-domain proximal neurodynamics with explicit stability and convergence guarantees for CMVIs.

Motivated by this gap, we propose a complex-valued proximal neural network (CPNN) for CMVIs. We prove existence and uniqueness and derive residual-based error bounds. We further establish global exponential stability for the continuous-time dynamics and linear convergence for a discrete-time implementation. In addition, CMVIs offer a convenient modeling framework for complex-valued equilibrium problems and inverse problems where both magnitude and phase are essential and structure-promoting nonsmooth regularization is often required, such as complex-valued MRI reconstruction (Bi et al., 2025), sparse recovery in complex signal processing (Zhang et al., 2021), and coherent imaging systems (Cetin et al., 2006). These practical considerations further motivate developing solver architectures that operate directly in the complex-domain.

The main contributions of this paper are summarized as follows: (1) We establish the existence and uniqueness of solutions to the CMVIs under suitable conditions, and derive an explicit error bound for the CMVIs. (2) We propose a CPNN method for solving CMVIs, and prove that the equilibrium points of the proposed network coincide with the solutions to the CMVIs. We also establish the well-posedness of the proposed network under mild conditions. (3) We establish global exponential stability of the continuous-time CPNN by combining Lyapunov analysis with the complex chain rule, and further prove linear convergence of its discrete-time implementation. (4) We demonstrate the effectiveness of the proposed CPNN through numerical experiments.

The paper is organized as follows. Section 2 presents preliminaries and basic lemmas. Section 3 establishes the existence and uniqueness of the CMVIs solutions and derives an associated error bound via complex-valued proximal mapping. Section 4 proposes the CPNN method, proves the equivalence between equilibrium points and CMVIs solutions, establishes well-posedness of the CPNN, and analyzes its global exponential stability under continuous-time and linear convergence in a discrete-time implementation. Section 5 presents numerical experiments to demonstrate the effectiveness of the proposed method. Section 6 concludes the paper and discusses possible future work.

2. Basic concepts and notations

In this paper, we denote by $i = \sqrt{-1}$ the imaginary unit, $z = x + iy \in \mathbb{C}^n$ and by x^T the transpose for x , where $x, y \in \mathbb{R}^n$, $\mathbb{R}^n$ and $\mathbb{C}^n$ are the n -dimensional real number field and n -dimensional complex number field, respectively. All vectors in $\mathbb{R}^n$ and $\mathbb{C}^n$ are considered as column vectors. A set $\Omega \subseteq \mathbb{C}^n$ is called convex if for any two points $z_1, z_2 \in \Omega$ and any t such that $0 \leq t \leq 1$, we have $tz_1 + (1-t)z_2 \in \Omega$. Denote by $[z_1, z_2] = \{z : tz_1 + (1-t)z_2, 0 \leq t \leq 1\}$ and denote by A^H the conjugate transpose of a matrix $A \in \mathbb{C}^{n \times n}$. We denote by $\bar{z}$ the conjugate and by z^H the conjugate transpose for $z \in \mathbb{C}^n$. Let $z(t) = x(t) + iy(t)$, where $z : \mathbb{R} \rightarrow \mathbb{C}^n$ and $x, y :$

$\mathbb{R} \rightarrow \mathbb{R}^n$. We denote by $\dot{x}(t) = \frac{dx}{dt}$, $\dot{z} = \frac{dz}{dt} = \dot{x}(t) + i\dot{y}(t)$. We denote by $\|z\|$ the Euclidean 2-norm of $z \in \mathbb{C}^n$, by $\text{Re } z$ the real part and by $\delta_\Omega(z)$ the indicator function on $\Omega \subseteq \mathbb{C}^n$, i.e., if $z \in \Omega$, $\delta_\Omega(z) = 0$ and $+\infty$ otherwise. We define the real inner product as follows:

$$(u, v)_{\mathbb{R}} = \text{Re}(u^H v) = x_1^T x_2 + y_1^T y_2, \quad (2.1)$$

where $u = x_1 + iy_1, v = x_2 + iy_2 \in \mathbb{C}^n$. A function $\Phi : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is said to be proper, if it satisfies: $\exists z_0 \in \mathbb{C}^n$ such that $\Phi(z_0) < +\infty$.

Definition 2.1. (Zheng et al., 2025) Let $\Omega \subseteq \mathbb{C}^n$ be a convex set. The function $g : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is said to be a convex function on Ω if for any two points $z_1, z_2 \in \Omega$ and $\lambda \in [0, 1]$, we have

$$g(\lambda z_1 + (1-\lambda)z_2) \leq \lambda g(z_1) + (1-\lambda)g(z_2).$$

Definition 2.2. (Zheng et al., 2025) Let $\Omega \subseteq \mathbb{C}^n$ be a nonempty subset and $\Phi : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$. A mapping $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is said to be

(i) α -strongly monotone on Ω if there is $\alpha > 0$ such that

$$\text{Re}((f(z_1) - f(z_2))^H (z_1 - z_2)) \geq \alpha \|z_1 - z_2\|^2, \quad \forall z_1, z_2 \in \Omega.$$

(ii) θ - Φ -strongly pseudomonotone on Ω if there is $\theta > 0$ such that for any two points $z_1, z_2 \in \Omega$,

$$\begin{aligned} \text{Re}(f(z_1)^H (z_2 - z_1)) + \Phi(z_2) - \Phi(z_1) &\geq 0 \\ \Rightarrow \text{Re}(f(z_2)^H (z_2 - z_1)) + \Phi(z_2) - \Phi(z_1) &\geq \theta \|z_1 - z_2\|^2. \end{aligned}$$

(iii) L -Lipschitz continuous on Ω if there is $L > 0$ such that for any two points $z_1, z_2 \in \Omega$,

$$\|f(z_1) - f(z_2)\| \leq L \|z_1 - z_2\|.$$

f is said to be α -strongly monotone if it is α -strongly monotone on $\mathbb{C}^n$, and so forth.

Proposition 2.1. Let $w, v, u \in \mathbb{C}^n$. Then

- (i) $\|w + v\|^2 = \|w\|^2 + \|v\|^2 + 2\text{Re}(w^H v)$.
- (ii) $\text{Re}(w^H v) \leq \|w\| \|v\|$.
- (iii) $2\text{Re}((w - u)^H (v - u)) = \|w - u\|^2 + \|v - u\|^2 - \|w - v\|^2$.
- (iv) $\text{Re}(u_1^H v) + \text{Re}(u_2^H v) = \text{Re}((u_1 + u_2)^H v)$.

Proof. (i)-(iii). It follows immediately from Zheng et al. (2025).

(iv). Let $u_1 = x_1 + iy_1, u_2 = x_2 + iy_2, v = v_1 + iv_2$. From (2.1), we have

$$\text{Re}(u_1^H v) = x_1^T v_1 + y_1^T v_2, \quad \text{Re}(u_2^H v) = x_2^T v_1 + y_2^T v_2$$

and consequently,

$$\text{Re}(u_1^H v) + \text{Re}(u_2^H v) = (x_1 + x_2)^T v_1 + (y_1 + y_2)^T v_2 = \text{Re}((u_1 + u_2)^H v).$$

This completes the proof. $\square$

Definition 2.3. Let $\Phi : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper convex function. The proximal mapping of Φ with parameter $\eta > 0$ is defined as follows:

$$\text{prox}_{\eta\Phi}(z) = \arg\min_{y \in \mathbb{C}^n} \left\{ \eta\Phi(y) + \frac{1}{2} \|y - z\|^2 \right\}. \quad (2.2)$$

Remark 2.1. For $\Phi(z) = \lambda \|z\|_1 = \lambda \sum_{j=1}^n |z_j|$, the complex-valued proximal mapping admits the closed-form elementwise radial shrinkage $\text{prox}_{\eta\Phi}(z)_j = (1 - \frac{\eta\lambda}{|z_j|})_+ z_j$, where $(a)_+ = \max\{a, 0\}$. However, by real-imaginary reformulation method, the same term becomes $\tilde{\Phi}(x, y) = \lambda \sum_{j=1}^n \sqrt{x_j^2 + y_j^2} = \lambda \sum_{j=1}^n \|(x_j, y_j)\|_2$, and its proximal mapping is equivalently given by the block soft-thresholding $\text{prox}_{\eta\tilde{\Phi}}(x_j, y_j) = \left(1 - \frac{\eta\lambda}{\sqrt{x_j^2 + y_j^2}}\right)_+ (x_j, y_j)$. Hence, the real-imaginary reformulation is mathematically equivalent and does not discard information. The distinction is that, in $\mathbb{R}^{2n}$, the proximal step is coupled on each two-dimensional block (x_j, y_j) and is no longer separable with respect to x_j and y_j individually. Thus, working directly in $\mathbb{C}^n$ yields a more compact and structurally consistent proximal implementation.

To define the derivative of complex variable function, [Qin, Feng, Song, Wen, and Xu \(2016\)](#) introduced an auxiliary function $f : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ of $g : \mathbb{C}^n \rightarrow \mathbb{R}$ as follows:

$$f(x, y) = g(x + iy) = g(z), \quad (2.3)$$

where $z = x + iy$, $x = \frac{z+\bar{z}}{2}$ and $y = \frac{-i(z-\bar{z})}{2}$. Then, g can be considered as a function of z and $\bar{z}$.

We first recall that the subdifferentials of the convex function $h : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ with respect to x and y are defined as

$$\partial_x h(x, y) = \{p \in \mathbb{R}^n : h(x, y) - h(s, y) \leq \langle p, x - s \rangle, \forall s \in \mathbb{R}^n\}$$

and

$$\partial_y h(x, y) = \{q \in \mathbb{R}^n : h(x, y) - h(x, t) \leq \langle q, y - t \rangle, \forall t \in \mathbb{R}^n\}.$$

Then, the subdifferential of g in the complex number field can be defined as follows.

Definition 2.4. ([Qin et al., 2016](#)) Let $g : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be a convex function. We denote by $\partial_c g(z)$ its subdifferential in the complex number field. The complex subdifferential of g at $z = x + iy \in \mathbb{C}^n$ is defined as

$$\partial_c g(z) := \partial_x f(x, y) + i\partial_y f(x, y),$$

where $f : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by (2.3).

Remark 2.2. The complex subdifferential provides the key tool to treat the nonsmooth regularizer Φ directly in $\mathbb{C}^n$. In particular, it yields the complex-domain optimality condition in [Lemma 2.4](#), which in turn enables us to establish nonexpansiveness of the complex-valued proximal mapping in [Lemma 2.6](#). These results are essential for our subsequent stability and convergence analysis without resorting to real-imaginary splitting.

Lemma 2.1. Assume that $g : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex and $z_1 \in \mathbb{C}^n$. Then, $\xi \in \partial_c g(z_1)$ if and only if

$$g(z_2) - g(z_1) \geq \operatorname{Re}(\xi^H(z_2 - z_1)), \quad \forall z_2 \in \mathbb{C}^n. \quad (2.4)$$

Proof. Necessity. It follows from [Qin et al. \(2016\)](#).

Sufficiency. Let $z_1 = x_1 + iy_1$ and $\xi = u + iv$. For any $z_2 = x_2 + iy_2 \in \mathbb{C}^n$, from (2.3) and (2.4) we obtain

$$g(z_2) - g(z_1) = f(x_2, y_2) - f(x_1, y_1) \geq \operatorname{Re}((u + iv)^H((x_2 - x_1) - i(y_2 - y_1))).$$

Setting $y_2 = y_1$ in the above inequality yields

$$f(x_2, y_1) - f(x_1, y_1) \geq \operatorname{Re}((u + iv)^H((x_2 - x_1) - i(y_1 - y_1))) = (u^T(x_2 - x_1)),$$

which yields that $u \in \partial_x f(x_1, y_1)$. Similarly, we have $v \in \partial_y f(x_1, y_1)$. Therefore

$$\xi = u + iv \in \partial_x f(x_1, y_1) + i\partial_y f(x_1, y_1) = \partial_c g(z_1).$$

This completes the proof. $\square$

Lemma 2.2. Assume that $g : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex. Then, for any $\xi_1 \in \partial_c g(z_1)$ and $\xi_2 \in \partial_c g(z_2)$, we have

$$\operatorname{Re}((\xi_1 - \xi_2)^H(z_1 - z_2)) \geq 0. \quad (2.5)$$

Proof. Since $\xi_1 \in \partial_c g(z_1)$ and $\xi_2 \in \partial_c g(z_2)$, from (2.4) we have

$$g(u) - g(z_1) \geq \operatorname{Re}(\xi_1^H(u - z_1)), \quad \forall u \in \mathbb{C}^n$$

and

$$g(v) - g(z_2) \geq \operatorname{Re}(\xi_2^H(v - z_2)), \quad \forall v \in \mathbb{C}^n.$$

Setting $u = z_2$ and $v = z_1$ in above inequalities yields that

$$g(z_2) - g(z_1) \geq \operatorname{Re}(\xi_1^H(z_2 - z_1)) \quad (2.6)$$

and

$$g(z_1) - g(z_2) \geq \operatorname{Re}(\xi_2^H(z_1 - z_2)). \quad (2.7)$$

Combining (2.6) and (2.7), we obtain

$$0 \geq \operatorname{Re}((\xi_1 - \xi_2)^H(z_2 - z_1)),$$

proving (2.5). This completes the proof. $\square$

According to [Qin et al. \(2016\)](#), we define $\mathbb{C}^n$ as an isometric isomorphism of $\mathbb{R}^{2n}$ by

$$z = x_1 + iy_1 \in \mathbb{C}^n \longleftrightarrow \tilde{z} = (x_1, y_1) \in \mathbb{R}^n \times \mathbb{R}^n = \mathbb{R}^{2n}$$

and

$$\|z\|^2 = x_1^2 + y_1^2 = \|\tilde{z}\|^2.$$

Therefore, for a function $h : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$, we can define $\tilde{h} : \mathbb{R}^{2n} \rightarrow \mathbb{R} \cup \{+\infty\}$ by $\tilde{h}(\tilde{z}) = h(z)$, where $\tilde{z}$ and z are as above. Hence, we obtain the following proposition.

Proposition 2.2. Let $f, g : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex functions and $a, b \geq 0$. Then, we have the following:

- (i) $h = af + bg$ is convex.
- (ii) $\tilde{f}$ is convex.
- (iii) $(f + g) = \tilde{f} + \tilde{g}$.

Proof. (i) Take arbitrary $z, w \in \mathbb{C}^n$ and $\lambda \in [0, 1]$. Since f and g are convex, we have

$$f(\lambda z + (1 - \lambda)w) \leq \lambda f(z) + (1 - \lambda)f(w)$$

and

$$g(\lambda z + (1 - \lambda)w) \leq \lambda g(z) + (1 - \lambda)g(w).$$

Multiplying the first inequality by $a \geq 0$ and the second by $b \geq 0$, and adding them together allows that

$$\begin{aligned} h(\lambda z + (1 - \lambda)w) &= af(\lambda z + (1 - \lambda)w) + bg(\lambda z + (1 - \lambda)w) \\ &\leq \lambda(af(z) + bg(z)) + (1 - \lambda)(af(w) + bg(w)) \\ &= \lambda h(z) + (1 - \lambda)h(w). \end{aligned}$$

This shows that h is convex.

(ii) It follows immediately from [Zhang et al. \(2015b\)](#).

(iii) For any $z = z_1 + iz_2 \in \mathbb{C}^n$ and $\tilde{z} = (z_1, z_2) \in \mathbb{R}^{2n}$, we have

$$\widetilde{(f + g)}(\tilde{z}) = (f + g)(z) = f(z) + g(z) = \tilde{f}(\tilde{z}) + \tilde{g}(\tilde{z}).$$

This completes the proof. $\square$

Lemma 2.3. Assume that $f, g : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ are convex. Suppose that at least one of f or g is continuous at a point $z \in \mathbb{C}^n$. Then, $\partial_c f(z) + \partial_c g(z) = \partial_c(f + g)(z)$.

Proof. We preliminarily prove $\partial_c f(z) + \partial_c g(z) \subseteq \partial_c(f + g)(z)$. Let $u \in \partial_c f(z)$ and $v \in \partial_c g(z)$. For any $y \in \mathbb{C}^n$, from (2.4) we have

$$f(y) \geq f(z) + \operatorname{Re}(u^H(y - z))$$

and

$$g(y) \geq g(z) + \operatorname{Re}(v^H(y - z)).$$

Adding these inequalities, and from [Proposition 2.1](#) (iv) we obtain

$$f(y) + g(y) \geq f(z) + g(z) + \operatorname{Re}((u + v)^H(y - z)),$$

which shows that

$$u + v \in \partial_c(f + g)(z).$$

That shows that

$$\partial_c f(z) + \partial_c g(z) \subseteq \partial_c(f + g)(z). \quad (2.8)$$

Next, we prove $\partial_c(f + g)(z) \subseteq \partial_c f(z) + \partial_c g(z)$. We first show that for a convex function $h : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$, we have

$$\xi \in \partial_c h(z) \Leftrightarrow \tilde{\xi} \in \partial \tilde{h}(\tilde{z}). \quad (2.9)$$

To this aim, let $\xi \in \partial_c h(z)$. Then, for any $y \in \mathbb{C}^n$ we have

$$\begin{aligned} h(y) &\geq h(z) + \operatorname{Re}(\xi^H(y - z)) \\ &= h(z) + \operatorname{Re}((\xi_1 + i\xi_2)^H((y_1 + iy_2) - (z_1 + iz_2))) \\ &= h(z) + \xi_1^T(y_1 - z_1) + \xi_2^T(y_2 - z_2), \end{aligned} \quad (2.10)$$

where $\xi = \xi_1 + i\xi_2, y = y_1 + iy_2, z = z_1 + iz_2 \in \mathbb{C}^n$. It is easy to see that (2.10) is equivalent to

$$\tilde{h}(\tilde{y}) \geq \tilde{h}(\tilde{z}) + (\tilde{\xi}^T(\tilde{y} - \tilde{z})),$$

which implies that (2.9) holds.

Since at least one of f or g is continuous at a point $z \in \mathbb{C}^n$, we obtain at least one of $\tilde{f}$ or $\tilde{g}$ is continuous at a point $\tilde{z} \in \mathbb{R}^{2n}$. According to Theorem 23.8 in Rockafellar (1970), we have

$$\partial(\tilde{f} + \tilde{g})(\tilde{z}) = \partial\tilde{f}(\tilde{z}) + \partial\tilde{g}(\tilde{z}).$$

By Proposition 2.2 (i)-(iii), $(\tilde{f} + \tilde{g})$ is convex, and from (2.9) it follows that

$$\xi \in \partial_c(f + g)(z) \Leftrightarrow \tilde{\xi} = (\xi_1, \xi_2) \in \partial(\tilde{f} + \tilde{g})(\tilde{z}) = \partial\tilde{f}(\tilde{z}) + \partial\tilde{g}(\tilde{z}).$$

Let $\xi \in \partial_c(f + g)(z)$. From above equivalence there are $\tilde{u} \in \partial\tilde{f}(\tilde{z})$ and $\tilde{v} \in \partial\tilde{g}(\tilde{z})$ such that $\tilde{\xi} = \tilde{u} + \tilde{v}$. Then, we have $u \in \partial_c f(z)$ and $v \in \partial_c g(z)$ such that $\xi = u + v$. This yields that

$$\partial_c(f + g)(z) \subseteq \partial_c f(z) + \partial_c g(z). \quad (2.11)$$

From (2.8) and (2.11) we obtain

$$\partial_c(f + g)(z) = \partial_c f(z) + \partial_c g(z).$$

This completes the proof. $\square$

Using Lemma 2.1, we have the following result.

Lemma 2.4. Assume that $g : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex. Then, $z_1 \in \mathbb{C}^n$ is a minimum point of g if and only if $0 \in \partial_c g(z_1)$.

Proof. Observe that from (2.4), one has

$$\begin{aligned} g(z_2) &\geq g(z_1), \forall z_2 \in \mathbb{C}^n \\ &\Leftrightarrow g(z_2) \geq g(z_1) + \operatorname{Re}(0^H(z_2 - z_1)), \forall z_2 \in \mathbb{C}^n \\ &\Leftrightarrow 0 \in \partial_c g(z_1). \end{aligned}$$

This completes the proof. $\square$

Lemma 2.5. Assume that $\eta > 0$, $z \in \mathbb{C}^n$ and $\Phi : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is a proper convex function. Then, $z_1 = \operatorname{prox}_{\eta\Phi}(z)$ if and only if

$$\operatorname{Re}((z_1 - z)^H(u - z_1)) \geq \eta\Phi(z_1) - \eta\Phi(u), \forall u \in \mathbb{C}^n.$$

Proof. Necessity. By the definition of the complex-valued proximal mapping, we have

$$z_1 = \operatorname{prox}_{\eta\Phi}(z) = \operatorname{argmin}_{y \in \mathbb{C}^n} \left\{ \eta\Phi(y) + \frac{1}{2} \|y - z\|^2 \right\}.$$

From Lemma 2.4 it follows that

$$0 \in \partial_c \Phi(z_1) + \frac{1}{\eta}(z_1 - z)$$

and so

$$\frac{1}{\eta}(z - z_1) \in \partial_c \Phi(z_1).$$

Applying Lemma 2.1, we have

$$\Phi(u) \geq \Phi(z_1) + \operatorname{Re}\left(\frac{1}{\eta}(z - z_1)^H(u - z_1)\right), \forall u \in \mathbb{C}^n,$$

or equivalently

$$\eta\Phi(u) \geq \eta\Phi(z_1) + \operatorname{Re}((z - z_1)^H(u - z_1)), \forall u \in \mathbb{C}^n.$$

As a consequence

$$\operatorname{Re}((z_1 - z)^H(u - z_1)) \geq \eta\Phi(z_1) - \eta\Phi(u), \forall u \in \mathbb{C}^n.$$

Sufficiency. Consider the function

$$F(u) = \eta\Phi(u) + \frac{1}{2} \|u - z\|^2, u \in \mathbb{C}^n.$$

Since Φ and $\frac{1}{2} \|\cdot - z\|^2$ are convex, from Proposition 2.2 (i) we have F is convex. Observe that $\frac{1}{2} \|\cdot - z\|^2$ is continuous. From Lemma 2.3 it follows that

$$\partial_c F(z_1) = \eta\partial_c \Phi(z_1) + (z_1 - z)$$

Since

$$\operatorname{Re}((z_1 - z)^H(u - z_1)) \geq \eta(\Phi(z_1) - \Phi(u)),$$

that is

$$\Phi(u) \geq \Phi(z_1) + \operatorname{Re}\left(\frac{1}{\eta}(z - z_1)^H(u - z_1)\right),$$

from Lemma 2.1 it follows that $\frac{1}{\eta}(z - z_1) \in \partial_c \Phi(z_1)$. As a consequence

$$0 \in \partial_c \eta\Phi(z_1) + (z_1 - z) = \partial_c F(z_1),$$

which implies $z_1 = \arg \min_{y \in \mathbb{C}^n} F(y) = \arg \min_{y \in \mathbb{C}^n} \left\{ \eta\Phi(y) + \frac{1}{2} \|y - z\|^2 \right\} = \operatorname{prox}_{\eta\Phi}(z)$, in view of Lemma 2.3. $\square$

Lemma 2.6. Assume that $\eta > 0$ and $\Phi : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is a proper convex function. Then, the complex-valued proximal mapping $\operatorname{prox}_{\eta\Phi}(\cdot)$ defined in (2.2) satisfies the following inequality:

$$\|\operatorname{prox}_{\eta\Phi}(u) - \operatorname{prox}_{\eta\Phi}(v)\| \leq \|u - v\|, \forall u, v \in \mathbb{C}^n.$$

Proof. Let $u_1 = \operatorname{prox}_{\eta\Phi}(u)$ and $v_1 = \operatorname{prox}_{\eta\Phi}(v)$, where $u, v, u_1, v_1 \in \mathbb{C}^n$. Similar to the proof of sufficiency in Lemma 2.5, from Lemma 2.4 it follows that

$$0 \in \partial_c \eta\Phi(u_1) + (u_1 - u), 0 \in \partial_c \eta\Phi(v_1) + (v_1 - v).$$

Let $u_2 = u - u_1 \in \partial_c \eta\Phi(u_1)$ and $v_2 = v - v_1 \in \partial_c \eta\Phi(v_1)$. From Lemma 2.2 it follows that

$$\operatorname{Re}(((u - u_1) - (v - v_1))^H(u_1 - v_1)) = \operatorname{Re}((u_2 - v_2)^H(u_1 - v_1)) \geq 0.$$

By Proposition 2.1 (iv) we have

$$\operatorname{Re}((u - v)^H(u_1 - v_1)) \geq \operatorname{Re}((u_1 - v_1)^H(u_1 - v_1)) = \|u_1 - v_1\|^2. \quad (2.12)$$

By Proposition 2.1 (ii) we also have

$$\operatorname{Re}((u - v)^H(u_1 - v_1)) \leq \|u - v\| \|u_1 - v_1\|. \quad (2.13)$$

Combining (2.12) and (2.13), we obtain

$$\|u_1 - v_1\|^2 \leq \|u - v\| \|u_1 - v_1\|$$

and so

$$\|\operatorname{prox}_{\eta\Phi}(u) - \operatorname{prox}_{\eta\Phi}(v)\| = \|u_1 - v_1\| \leq \|u - v\|.$$

This completes the proof. $\square$

In this paper, we consider the following complex-valued mixed variational inequality problems (CMVIs): Find $z \in \mathbb{C}^n$ such that it satisfies the following inequality

$$\operatorname{Re}(f(z)^H(u - z)) + \Phi(u) - \Phi(z) \geq 0, \forall u \in \mathbb{C}^n, \quad (2.14)$$

where $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$ and $\Phi : \mathbb{C}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is a proper convex function. In particular, if Φ is a constant function, the inequality (2.14) reduces to a complex-valued variational inequality as shown in Zheng et al. (2025).

Remark 2.3. Following Facchinei and Pang (2003), one can deduce that the CMVIs (2.14) admit a solution under suitable conditions. Therefore, in this paper, we assume that the CMVIs(2.14) admits a solution.

3. Error bounds of the unique solution of the CMVIs

In this section, we prove error bounds of the unique solution of the CMVIs(2.14) by using complex-valued proximal mapping under suitable assumptions.

Theorem 3.1. Assume that $\eta > 0$ and $z^* \in \mathbb{C}^n$. $z^* = \operatorname{prox}_{\eta\Phi}(z^* - \eta f(z^*))$ if and only if z^* is a solution of the CMVIs(2.14).

Proof. We observe that

$$z^* = \operatorname{prox}_{\eta\Phi}(z^* - \eta f(z^*))$$

$$\Leftrightarrow \operatorname{Re}((z^* - z^* + \eta f(z^*))^H(y - z^*)) \geq \eta\Phi(z^*) - \eta\Phi(y), \forall y \in \mathbb{C}^n$$

$$\Leftrightarrow \operatorname{Re}(\eta f(z^*)^H(y - z^*)) \geq \eta\Phi(z^*) - \eta\Phi(y), \forall y \in \mathbb{C}^n$$

$$\Leftrightarrow \operatorname{Re}(f(z^*)^H(y - z^*)) \geq \Phi(z^*) - \Phi(y), \forall y \in \mathbb{C}^n,$$

where the second equivalence follows from Lemma 2.5. $\square$

Theorem 3.2. Assume that $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is $\theta - \Phi$ -strongly pseudomonotone with a constant $\theta > 0$. Then, the CMVIs (2.14) has a unique solution.

Proof. Let $z_1^*, z_2^* \in \mathbb{C}^n$ be two solutions of the CMVIs (2.14). Then

$$\operatorname{Re}(f(z_1^*)^H(u - z_1^*)) + \Phi(u) - \Phi(z_1^*) \geq 0, \forall u \in \mathbb{C}^n.$$

Let $u = z_2^*$, we obtain

$$\operatorname{Re}(f(z_1^*)^H(z_2^* - z_1^*)) + \Phi(z_2^*) - \Phi(z_1^*) \geq 0.$$

Since f is $\theta - \Phi$ -strongly pseudomonotone, it follows that

$$\operatorname{Re}(f(z_2^*)^H(z_2^* - z_1^*)) + \Phi(z_2^*) - \Phi(z_1^*) \geq \theta \|z_2^* - z_1^*\|^2. \quad (3.1)$$

Similarly, since z_2^* is a solution of the CMVIs (2.14) we obtain

$$\operatorname{Re}(f(z_2^*)^H(z_1^* - z_2^*)) + \Phi(z_1^*) - \Phi(z_2^*) \geq 0. \quad (3.2)$$

Combining (3.1) and (3.2), from Proposition 2.1 (iv) we have

$$0 \geq \theta \|z_2^* - z_1^*\|^2$$

and so $z_1^* = z_2^*$. This proves that the CMVIs (2.14) has a unique solution. $\square$

Theorem 3.3. Assume that $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is $\theta - \Phi$ -strongly pseudomonotone with a constant $\theta > 0$ and L -Lipschitz continuous with $L > 0$. If z^* is a solution of the CMVIs (2.14) and $\eta L^2 < 2\theta$, then

$$\|w - z^*\| \leq \frac{1}{\sqrt{1 + 2\eta\theta - \eta^2 L^2}} \|z - z^*\|, \forall z \in \mathbb{C}^n$$

where $w = \operatorname{prox}_{\eta\Phi}(z - \eta f(z))$.

Proof. By Lemma 2.5, $w = \operatorname{prox}_{\eta\Phi}(z - \eta f(z))$ if and only if

$$\operatorname{Re}((w - z + \eta f(z))^H(y - w)) \geq \eta\Phi(w) - \eta\Phi(y), \forall y \in \mathbb{C}^n. \quad (3.3)$$

Substituting $y = z^*$ in (3.3) yields

$$\operatorname{Re}((w - z + \eta f(z))^H(z^* - w)) \geq \eta\Phi(w) - \eta\Phi(z^*),$$

and from Proposition 2.1 (iv) it follows that

$$\operatorname{Re}((z - w)^H(z^* - w)) \leq \eta[\operatorname{Re}(f(z)^H(z^* - w)) + \Phi(z^*) - \Phi(w)]. \quad (3.4)$$

Since z^* is a solution of the CMVIs (2.14), we have

$$\operatorname{Re}(f(z^*)^H(w - z^*)) + \Phi(w) - \Phi(z^*) \geq 0.$$

Since $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is $\theta - \Phi$ -strongly pseudomonotone, we obtain

$$\operatorname{Re}(f(w)^H(w - z^*)) + \Phi(w) - \Phi(z^*) \geq \theta \|w - z^*\|^2. \quad (3.5)$$

As a consequence

$$\begin{aligned} & 2\eta[\operatorname{Re}(f(z)^H(z^* - w)) + \Phi(z^*) - \Phi(w)] \\ &= 2\eta \operatorname{Re}((f(z) - f(w))^H(z^* - w)) - 2\eta[\operatorname{Re}(f(w)^H(w - z^*)) + \Phi(w) - \Phi(z^*)] \\ &\leq 2\eta\|f(z) - f(w)\|\|z^* - w\| - 2\eta\theta\|w - z^*\|^2 \\ &\leq 2\eta L\|z - w\|\|z^* - w\| - 2\eta\theta\|w - z^*\|^2 \\ &\leq \|z - w\|^2 + \eta^2 L^2\|z^* - w\|^2 - 2\eta\theta\|w - z^*\|^2 \\ &= \|z - w\|^2 + (\eta^2 L^2 - 2\eta\theta)\|w - z^*\|^2, \end{aligned} \quad (3.6)$$

where the first inequality follows from (3.5) and Proposition 2.1 (ii) and the second inequality follows from the L -Lipschitz continuity of the mapping f . Combining (3.4) and (3.6), we obtain

$$2 \operatorname{Re}((z - w)^H(z^* - w)) \leq \|z - w\|^2 + (\eta^2 L^2 - 2\eta\theta)\|w - z^*\|^2. \quad (3.7)$$

From Proposition 2.1(iii) it follows that

$$2 \operatorname{Re}((z - w)^H(z^* - w)) = \|z - w\|^2 + \|z^* - w\|^2 - \|z - z^*\|^2. \quad (3.8)$$

Combining (3.7) and (3.8) we obtain

$$(1 + 2\eta\theta - \eta^2 L^2)\|w - z^*\|^2 \leq \|z - z^*\|^2$$

and from the fact that $\eta L^2 < 2\theta$ it follows that

$$\|w - z^*\| \leq \frac{1}{\sqrt{1 + 2\eta\theta - \eta^2 L^2}} \|z - z^*\|.$$

This completes the proof. $\square$

Theorem 3.4. Assume that $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is $\theta - \Phi$ -strongly pseudomonotone with a constant $\theta > 0$ and L -Lipschitz continuous with $L > 0$. If $\eta L^2 < 2\theta$, the following properties hold

$$\|z - z^*\| \leq \frac{1}{1 - \frac{1}{\sqrt{1 + 2\eta\theta - \eta^2 L^2}}} \|z - \operatorname{prox}_{\eta\Phi}(z - \eta f(z))\|, \forall z \in \mathbb{C}^n.$$

and

$$\|z - z^*\| \geq \frac{1}{2} \|z - \operatorname{prox}_{\eta\Phi}(z - \eta f(z))\|, \forall z \in \mathbb{C}^n.$$

Proof. Let $w = \operatorname{prox}_{\eta\Phi}(z - \eta f(z))$, for $z \in \mathbb{C}^n$. Then from the fact that $\eta L^2 < 2\theta$ it follows that

$$\begin{aligned} \operatorname{Re}((z - z^*)^H(z - w)) &= \|z - z^*\|^2 + \operatorname{Re}((z - z^*)^H(z^* - w)) \\ &= \|z - z^*\|^2 - \operatorname{Re}((z - z^*)^H(w - z^*)) \\ &\geq \|z - z^*\|^2 - \|z - z^*\|\|z^* - w\| \\ &\geq \|z - z^*\|^2 - \frac{1}{\sqrt{1 + 2\eta\theta - \eta^2 L^2}} \|z - z^*\|^2 \\ &= (1 - \frac{1}{\sqrt{1 + 2\eta\theta - \eta^2 L^2}}) \|z - z^*\|^2, \forall z \in \mathbb{C}^n, \end{aligned} \quad (3.9)$$

where the first inequality follows from Proposition 2.1 (ii) and the second inequality follows from Theorem 3.3. Now, from Proposition 2.1 (ii) we have

$$\operatorname{Re}((z - z^*)^H(z - w)) \leq \|z - z^*\|\|z - w\|. \quad (3.10)$$

Combining (3.9) and (3.10), one gets

$$\|z - z^*\| \leq \frac{1}{1 - M} \|z - w\|, \forall z \in \mathbb{C}^n,$$

where $M = \frac{1}{\sqrt{1 + 2\eta\theta - \eta^2 L^2}}$. Since $\eta L^2 < 2\theta$, again from Proposition 2.1 (iii) and Theorem 3.3 we have

$$\begin{aligned} 2 \operatorname{Re}((z - w)^H(z - z^*)) &= \|z - w\|^2 + \|z - z^*\|^2 - \|w - z^*\|^2 \\ &\geq \|z - w\|^2 + \|z - z^*\|^2 - (\frac{1}{1 + 2\eta\theta - \eta^2 L^2}) \|z - z^*\|^2 \\ &= \|z - w\|^2 + (1 - \frac{1}{1 + 2\eta\theta - \eta^2 L^2}) \|z - z^*\|^2 \\ &\geq \|z - w\|^2, \forall z \in \mathbb{C}^n. \end{aligned} \quad (3.11)$$

Combining Proposition 2.1 (ii) and (3.11) we have

$$\|z - z^*\| \geq \frac{1}{2} \|z - w\|, \forall z \in \mathbb{C}^n.$$

This completes the proof. $\square$

4. Complex-valued proximal neural network method for solving CMVIs

In this section, we propose a CPNN method for solving CMVIs by using the complex-valued proximal mapping, prove the equivalence between the equilibrium points and the solutions of the CMVIs(2.14), demonstrate that this CPNN has a unique solution, establish the global exponential stability of the CPNN, and analyze the linear convergence of the discrete CPNN. We consider the following CPNN system for solving CMVIs(2.14):

$$\dot{z} = \frac{dz}{dt} = \beta(-z(t) + \operatorname{prox}_{\eta\Phi}(z(t) - \eta f(z(t)))), \quad (4.1)$$

where $\eta > 0$, $\beta > 0$, $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$; $z : \mathbb{R} \rightarrow \mathbb{C}^n$ with $z(0) = z_0 \in \mathbb{C}^n$. To consider the CPNN (4.1) has the unique solution $z(t)$, we extend Theorem 2.1 in Perko (2013) to $\mathbb{C}^n$.

Definition 4.1. (Global exponential stability in $\mathbb{C}^n$) Consider the dynamical system:

$$\dot{z} = \varphi(z(t)), z(0) = z_0 \in \mathbb{C}^n, \quad (4.2)$$

where $\varphi : \mathbb{C}^n \rightarrow \mathbb{C}^n$ and $z : \mathbb{R} \rightarrow \mathbb{C}^n$. A point z^* is called an equilibrium point of (4.2) if $\varphi(z^*) = 0$. The equilibrium point z^* is said to be globally

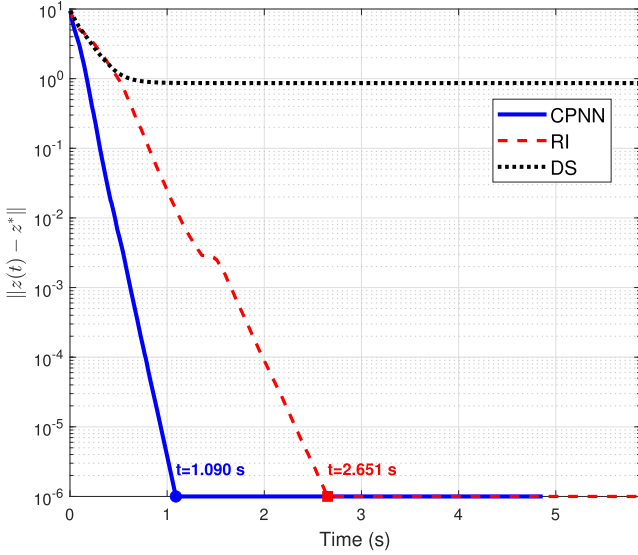


Fig. 1. Trajectories of the reconstruction error $\|z(t) - z^*\|$ versus time for three methods (CPNN, real isomorphism (RI), and direct separation (DS)) in [Example 5.1](#), shown on a semilog scale ($m = 500$, $n = 1000$, $\beta = 25$, $\lambda = 0.1$, $\eta = 0.95$, sparsity = 0.05).

exponentially stable if there exists a constant $\alpha > 0$ such that for every solution $z(t)$ with initial condition $z(0)$, the following holds

$$\|z(t) - z^*\| \leq \|z(0) - z^*\|e^{-\alpha t}, \quad \forall t > 0.$$

Lemma 4.1. ([Perko, 2013](#)) Assume that $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an L -Lipschitz continuous mapping with $L > 0$ and $x : \mathbb{R} \rightarrow \mathbb{R}^n$. For an initial point $x_0 \in \mathbb{R}^n$, the initial value problem

$$\dot{x}(t) = F(x(t)), \quad x(0) = x_0 \in \mathbb{R}^n$$

has a unique solution $x(t)$ for all $t > 0$.

Theorem 4.1. (Picard theorem for autonomous systems in $\mathbb{C}^n$) Assume that $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is an L -Lipschitz continuous mapping with $L > 0$ and $z : \mathbb{R} \rightarrow \mathbb{C}^n$. For an initial point $z_0 \in \mathbb{C}^n$, the initial value problem

$$\dot{z}(t) = F(z(t)), \quad z(0) = z_0 \in \mathbb{C}^n \quad (4.3)$$

has a unique solution $z(t)$ for all $t > 0$.

Proof. Let $z = x + iy$ and $F(z) = u(x, y) + iv(x, y)$, where $x, y \in \mathbb{R}^n$ and $u, v : \mathbb{R}^{2n} \rightarrow \mathbb{R}^n$. We define $\tilde{F} : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ by $\tilde{F}(\tilde{z}) := (u(x, y), v(x, y)) \in \mathbb{R}^{2n}$, where $\tilde{z} := (x, y) \in \mathbb{R}^{2n}$ and define $\tilde{z} : \mathbb{R} \rightarrow \mathbb{R}^{2n}$ by $\tilde{z}(t) = (x(t), y(t))$, where $x, y : \mathbb{R} \rightarrow \mathbb{R}^n$ and $z(t) = x(t) + iy(t)$. Then the system (4.3) can be transformed into the following system

$$\dot{\tilde{z}}(t) = \tilde{F}(\tilde{z}(t)), \quad \tilde{z}(0) = \tilde{z}_0 \in \mathbb{R}^{2n}. \quad (4.4)$$

Let $\tilde{z}_1 = (x_1, y_1)$ and $\tilde{z}_2 = (x_2, y_2)$, where $x_i, y_i \in \mathbb{R}^n (i = 1, 2)$, $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. Since F is L -Lipschitz continuous with $L > 0$, we have

$$\begin{aligned} \|\tilde{F}(\tilde{z}_1) - \tilde{F}(\tilde{z}_2)\|^2 &= \|(u(x_1, y_1), v(x_1, y_1)) - (u(x_2, y_2), v(x_2, y_2))\|^2 \\ &= \|F(z_1) - F(z_2)\|^2 \\ &\leq L^2 \|z_1 - z_2\|^2 \\ &= L^2 \|\tilde{z}_1 - \tilde{z}_2\|^2. \end{aligned}$$

Therefore, $\tilde{F}$ is L -Lipschitz continuous. According to [Lemma 4.1](#), we obtain the system (4.4) has a unique solution $\tilde{z}(t)$ for all $t > 0$ and so the system (4.3) has a unique solution $z(t)$ for all $t > 0$. This completes the proof. $\square$

Theorem 4.2. Let $z^* \in \mathbb{C}^n$. z^* is an equilibrium point of the CPNN (4.1) if and only if z^* is a solution of the CMVIs (2.14).

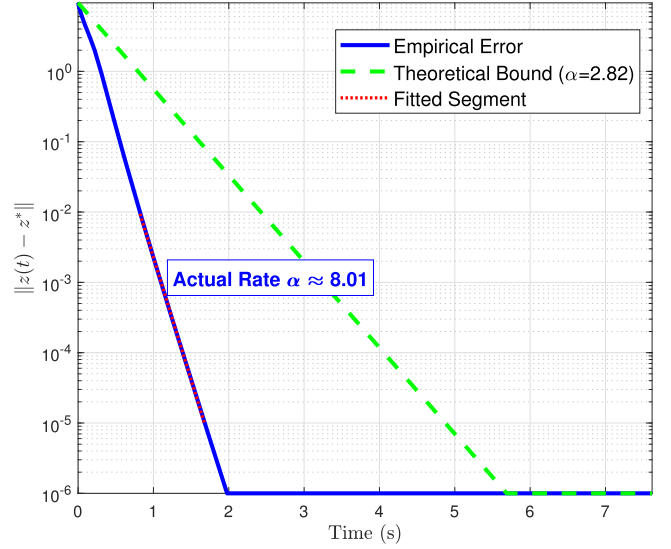


Fig. 2. The reconstruction error $\|z(t) - z^*\|$ of CPNN versus time in [Example 5.1](#), shown on a semilog scale ($m = 500$, $n = 1000$, $\beta = 25$, $\lambda = 0.1$, $\eta = 0.95$, sparsity = 0.05).

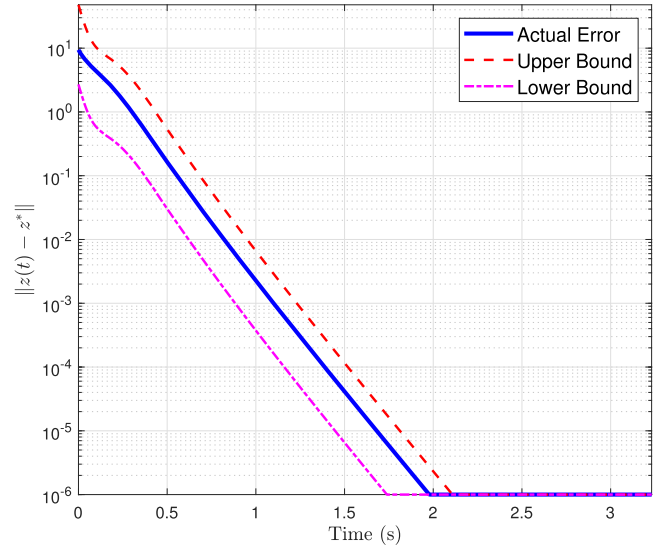


Fig. 3. Trajectories of $\|z(t) - z^*\|$ and the corresponding theoretical upper and lower bounds in [Theorem 3.4](#) versus time for CPNN in [Example 5.1](#), shown on a semilog scale ($m = 500$, $n = 1000$, $\beta = 25$, $\lambda = 0.1$, $\eta = 0.95$, sparsity = 0.05).

Proof. Since z^* is an equilibrium point of the CPNN (4.1), we obtain $z^* = \text{prox}_{\eta\Phi}(z^* - \eta f(z^*))$. According to [Theorem 3.1](#), z^* is a solution of the CMVIs (2.14). This completes the proof. $\square$

Theorem 4.3. Assume that $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is an L -Lipschitz continuous function with $L > 0$. Then, the CPNN (4.1) has a unique solution $z(t)$ for all $t > 0$.

Proof. Let $\eta > 0$. We construct the function $\psi : \mathbb{C}^n \rightarrow \mathbb{C}^n$ defined by

$$\psi(z) = -z + \text{prox}_{\eta\Phi}(z - \eta f(z)), \quad \forall z \in \mathbb{C}^n.$$

Then, for any $z_1, z_2 \in \mathbb{C}^n$, we have

$$\|\psi(z_1) - \psi(z_2)\| = \|-z_1 + z_2 + \text{prox}_{\eta\Phi}(z_1 - \eta f(z_1)) - \text{prox}_{\eta\Phi}(z_2 - \eta f(z_2))\|.$$

Since f is L -Lipschitz continuous and the proximal mapping is nonexpansive (by [Lemma 2.6](#)), we obtain

$$\|\psi(z_1) - \psi(z_2)\| \leq \|z_1 - z_2\| + \|\text{prox}_{\eta\Phi}(z_1 - \eta f(z_1)) - \text{prox}_{\eta\Phi}(z_2 - \eta f(z_2))\|$$

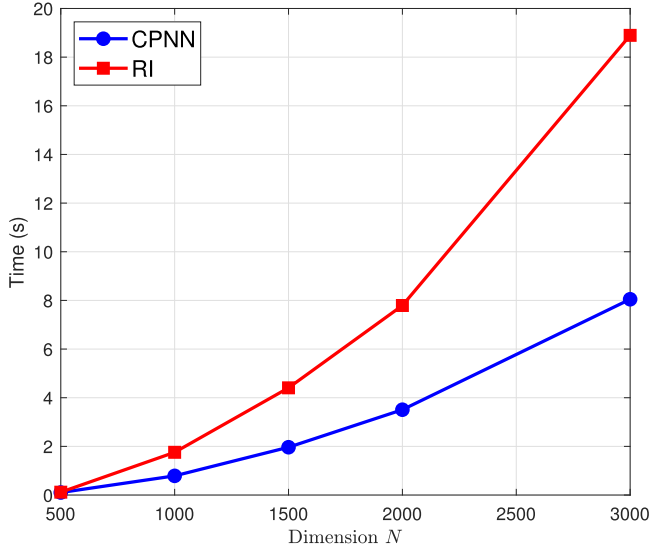


Fig. 4. Time (s) required to reach the prescribed convergence threshold versus the problem dimension N for CPNN and RI in Example 5.1 ($\beta = 25$, $\lambda = 0.1$, $\eta = 0.95$, $\text{sparsity} = 0.05$).

$$\begin{aligned}
 &\leq \|z_1 - z_2\| + \|z_1 - z_2 + \eta(f(z_2) - f(z_1))\| \\
 &\leq 2\|z_1 - z_2\| + \eta\|f(z_1) - f(z_2)\| \\
 &\leq (2 + \eta L)\|z_1 - z_2\|.
 \end{aligned}$$

We conclude that ψ is L_1 -Lipschitz continuous with $L_1 = \eta L + 2$. From Theorem 4.1 we obtain that the CPNN (4.1) has a unique solution $z(t)$ for all $t > 0$. $\square$

Lemma 4.2. (Liu et al., 2020) $f : \mathbb{C}^n \rightarrow \mathbb{R}$ is convex, $z(t)$ is absolutely continuous on each compact interval, we have

$$\frac{db(z(t))}{dt} = \text{Re}(\xi^H \dot{z}(t)), \quad \text{any } \xi \in \partial_c b(z(t)),$$

for a.e. $t \in [0, +\infty)$.

Based on the above-mentioned complex subdifferential chain rule and Wirtinger calculus, we obtain the following results.

Theorem 4.4. Assume that $f : \mathbb{C}^n \rightarrow \mathbb{R}$ is $\theta - \Phi$ -strongly pseudomonotone and L -Lipschitz continuous. Then, the CPNN (4.1) is globally exponentially stable at the equilibrium point z^* .

Proof. Consider the Lyapunov function $V : \mathbb{C}^n \rightarrow \mathbb{R}$ defined by

$$V(z) = \frac{1}{2} \|z - z^*\|^2, \quad z \in \mathbb{C}^n. \quad (4.5)$$

Computing the derivative of Lyapunov function defined by (4.5), it follows from the CPNN (4.1) that

$$\begin{aligned}
 \frac{dV}{dt} &= \text{Re}((z(t) - z^*)^H \dot{z}(t)) \\
 &= \beta \{ \text{Re}((z(t) - z^*)^H (-z(t) + w(t))) \} \\
 &= \beta \{ \text{Re}((z(t) - z^*)^H (-z(t) + z^* - z^* + w(t))) \} \\
 &= \beta \{ \text{Re}((z(t) - z^*)^H (w(t) - z^*)) - \|z(t) - z^*\|^2 \} \\
 &\leq \beta \{ \|z(t) - z^*\| \|w(t) - z^*\| - \|z(t) - z^*\|^2 \},
 \end{aligned} \quad (4.6)$$

where $w(t) = \text{prox}_{\eta\Phi}(z(t) - \eta f(z(t)))$. From Theorem 3.3 and (4.6) we have

$$\begin{aligned}
 \frac{dV}{dt} &\leq \frac{\beta}{\sqrt{1 + 2\eta\theta - \eta^2 L^2}} \|z(t) - z^*\|^2 - \beta \|z(t) - z^*\|^2 \\
 &= \left(\frac{\beta}{\sqrt{1 + 2\eta\theta - \eta^2 L^2}} - \beta \right) \|z(t) - z^*\|^2 \\
 &= -\alpha \|z(t) - z^*\|^2,
 \end{aligned}$$

where $\alpha = \beta - \frac{\beta}{\sqrt{1 + 2\eta\theta - \eta^2 L^2}} > 0$. This yields that

$$\frac{dV}{dt} \leq -2\alpha V(z(t)),$$

and by integration for all $t > 0$, we have

$$\|z(t) - z^*\| \leq \|z(0) - z^*\| e^{-\alpha t},$$

which proves that the CPNN (4.1) is globally exponentially stable at the equilibrium point z^* . $\square$

Corollary 4.1. Assume that $f : \mathbb{C}^n \rightarrow \mathbb{R}$ is α -strongly monotone and L -Lipschitz continuous. Then, the CPNN (4.1) is globally exponentially stable at the equilibrium point z^* .

Next, to study the linear convergence of the discrete-time CPNN (4.1), we discretize the continuous-time model CPNN (4.1) as follows:

$$z_{n+1} = (1 - \beta\psi_n)z_n + \beta\psi_n \text{prox}_{\eta\Phi}(z_n - \eta f(z_n)), \quad (4.7)$$

where $\beta, \eta, \psi_n > 0$, the initial point $z_0 \in \mathbb{C}^n$ and $\beta\psi_n \in (0, 1]$.

Theorem 4.5. Assume that $f : \mathbb{C}^n \rightarrow \mathbb{R}$ is $\theta - \Phi$ -strongly pseudomonotone and L -Lipschitz continuous. Let $h_n = \beta\psi_n$ be a step size sequence contained in $[h, 1]$ with a fixed lower bound $h > 0$, and let the initial point $z_0 \in \mathbb{C}^n$. If z^* a solution of the CMVIs (2.14) and $\eta L^2 < 2\theta$, then the sequence generated by the iterative scheme (4.7) converges linearly to a solution of the CMVIs (2.14).

Proof. Let $w_n = \text{prox}_{\eta\Phi}(z_n - \eta f(z_n))$ and $q = \frac{1}{1 + 2\eta\theta - \eta^2 L^2}$. Then from Theorem 3.3 we have

$$\|w_n - z^*\|^2 \leq q \|z_n - z^*\|^2.$$

As a consequence

$$\begin{aligned}
 \|z_{n+1} - z^*\|^2 &= \|(1 - h_n)z_n + h_n w_n - z^*\|^2 \\
 &= \|(1 - h_n)(z_n - z^*) + h_n(w_n - z^*)\|^2 \\
 &= (1 - h_n)\|z_n - z^*\|^2 + h_n\|w_n - z^*\|^2 - h_n(1 - h_n)\|z_n - w_n\|^2 \\
 &\leq (1 - h_n)\|z_n - z^*\|^2 + h_n q \|z_n - z^*\|^2 \\
 &\leq (1 - h_n(1 - q))\|z_n - z^*\|^2 \\
 &\leq (1 - h(1 - q))\|z_n - z^*\|^2.
 \end{aligned}$$

Since $\eta L^2 < 2\theta$, we obtain $q \in (0, 1)$ and so $(1 - h(1 - q)) \in (0, 1)$. Therefore, the sequence $\{z_n\}$ generated by the iterative scheme (4.7) converges linearly to the solution of the CMVIs (2.14). $\square$

Remark 4.1. The Lipschitz continuity and $\theta - \Phi$ strong pseudomonotonicity assumptions may be restrictive in some applications. However, in this paper, they are adopted as standard sufficient conditions to obtain global exponential stability (Theorem 4.4) and linear convergence of the discrete-time scheme (Theorem 4.5), built upon the residual error bound in Theorem 3.4. In structured cases, such constants can often be estimated or bounded in practice.

5. Numerical examples

In this section, numerical experiments are presented to verify the effectiveness of the proposed CPNN (4.1) for solving CMVIs (2.14). Specifically, we consider a high-dimensional complex-valued compressed sensing problem with $n = 1000$ and $m = 500$, using a partial DFT sensing matrix.

Example 5.1. Consider a complex compressed sensing problem

$$\min_{z \in \mathbb{C}^n} \frac{1}{2} \|Az - y\|^2 + \lambda \|z\|_1, \quad (5.1)$$

where $A \in \mathbb{C}^{m \times n}$ is a partial DFT sensing matrix, y is a measurement vector and $\lambda = 0.1$. We reformulate this problem as a CMVI(2.14), i.e., find $z \in \mathbb{C}^n$ such that

$$\text{Re}(f(z)^H(u - z)) + \Phi(u) - \Phi(z) \geq 0, \quad \forall u \in \mathbb{C}^n,$$

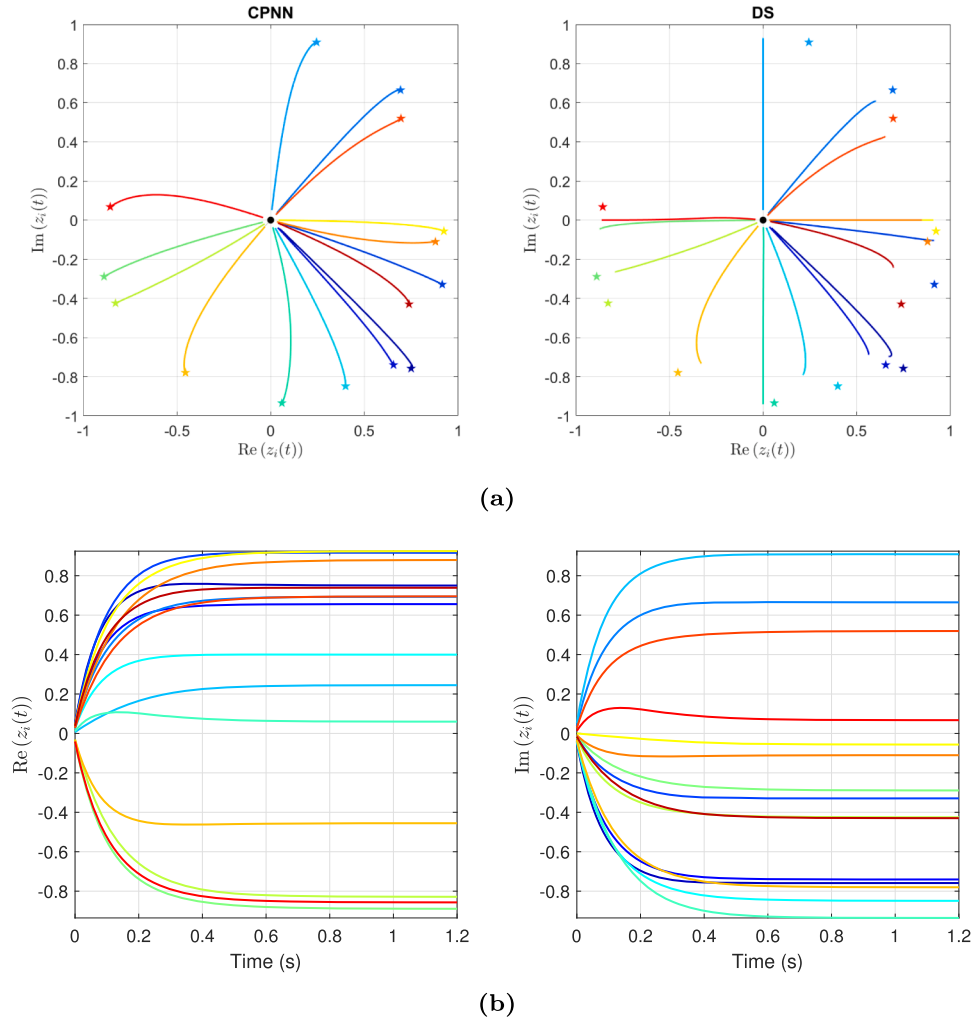


Fig. 5. Phase-plane trajectories of 15 randomly selected components $z_i(t)$ in the complex plane for CPNN and DS in [Example 5.1](#): (a) Complex-plane trajectories produced by CPNN and DS. (b) Time evolutions of $\text{Re}(z_i(t))$ and $\text{Im}(z_i(t))$ for the same 15 components under CPNN ($m = 500$, $n = 1000$, $\beta = 25$, $\lambda = 0.1$, $\eta = 0.95$, sparsity = 0.05).

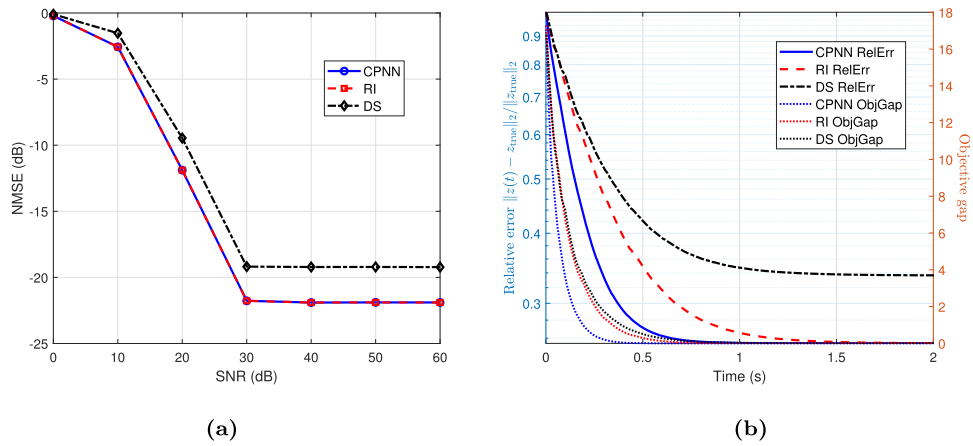


Fig. 6. Noise robustness in [Example 5.1](#): (a) Robustness under measurement noise: reconstruction NMSE (dB) versus SNR for CPNN, RI and DS. (b) Typical convergence trajectories at SNR = 20 dB: relative reconstruction error $\|z(t) - z_{\text{true}}\|_2 / \|z_{\text{true}}\|_2$ (left axis) and objective gap (right axis) versus time.

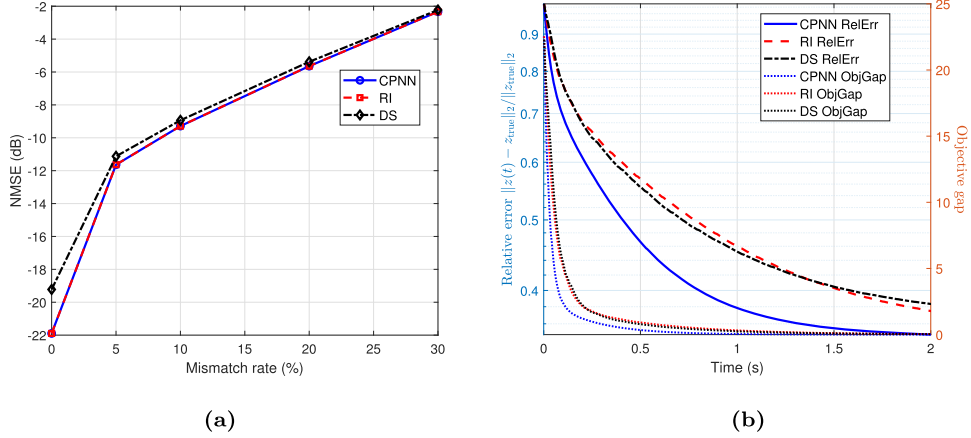


Fig. 7. Model mismatch robustness in [Example 5.1](#): (a) Robustness under different mismatch rate: ratesreconstruction NMSE (dB) versus mismatch rate for CPNN, RI and DS. (b) Typical convergence at a mismatch rate of 10%: relative error $\|z(t) - z_{\text{true}}\|_2 / \|z_{\text{true}}\|_2$ (left axis) and objective gap (right axis) versus time.

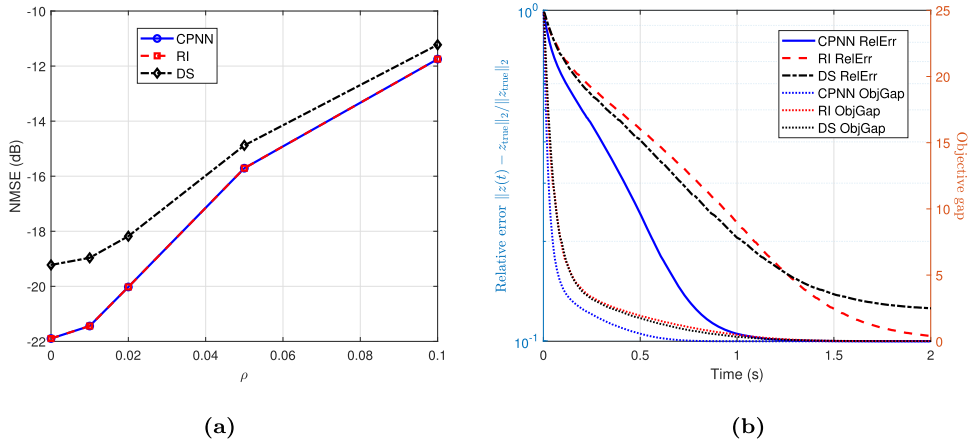


Fig. 8. Perturbation robustness in [Example 5.1](#): (a) Robustness under different perturbation levels: reconstruction NMSE (dB) versus the perturbation coefficient ρ for CPNN, RI and DS. (b) Typical convergence trajectories at a fixed perturbation level $\rho = 0.02$: relative reconstruction error $\|z(t) - z_{\text{true}}\|_2 / \|z_{\text{true}}\|_2$ (left axis) and objective gap (right axis) versus time.

where

$$f(z) = A^H(Az - y), \quad \Phi(z) = \lambda \|z\|_1.$$

It can be verified that $\Phi(z) = \lambda \|z\|_1$ is proper, convex, and lower semicontinuous, and that $f(z) = A^H(Az - y)$ is L -Lipschitz continuous with $L = \|A^H A\|_2 = \|A\|_2^2 = 1$ under the normalization used in our experiments. For the θ -related condition in [Theorem 4.4](#) and [4.5](#), the present compressed-sensing example is interpreted in a structured sparse-recovery sense. In particular, under the RIP-type condition associated with the partial DFT sensing matrix, this condition is reflected through the restricted strong monotonicity of the data-fidelity operator on sparse vectors. Therefore, the assumptions on Φ and the Lipschitz continuity of f are satisfied explicitly in this example, whereas the θ -related condition is understood through the restricted strong monotonicity induced by the sparse-recovery structure. Next, we will verify the effectiveness of the CPNN method in solving [Example 5.1](#). [Fig. 1](#) compares the reconstruction error trajectories of the proposed CPNN, the real-valued isomorphism method (RI), and the direct separation method (DS) with respect to time, where the vertical axis is displayed on a semilog scale. The results show that CPNN reaches the prescribed accuracy threshold within approximately 1.09 s and converges faster than both RI and DS. [Fig. 2](#) further illustrates the decay behavior of the CPNN reconstruction error on a semilog scale. The error curve appears approximately linear in the semilog coordinates, which is consistent with the

global exponential stability result. Moreover, the empirically fitted decay rate is $\alpha \approx 8.01$, which is larger than the conservative theoretical lower bound $\alpha \approx 2.82$, indicating that the theoretical estimate is conservative and that the method converges faster in practice. [Fig. 3](#) provides a numerical verification of [Theorem 3.4](#), showing that the observed error trajectory lies between the derived theoretical upper and lower bounds, supporting the validity of the error bound analysis. [Fig. 4](#), together with the numerical values reported in [Table 1](#), clearly illustrates the scalability advantage of CPNN. While the RI suffers from rapidly increasing computational cost due to the $2N$ -dimensional real-valued reformulation, CPNN directly operates in the complex-domain and avoids such overhead. Consequently, the convergence time of CPNN grows more moderately with respect to N , which makes it particularly well suited for large-scale complex-valued problems. [Fig. 5](#) shows the dynamic trajectories of 15 representative components. As shown in [Fig. 5\(a\)](#), the proposed CPNN exhibits smooth convergence and preserves coherent trajectories in the complex plane, whereas DS produces more irregular trajectories. Moreover, [Fig. 5\(b\)](#) shows that under CPNN both $\text{Re}(z_i(t))$ and $\text{Im}(z_i(t))$ rapidly converge to equilibrium point within the displayed time interval. These observations are consistent with the lower error floor in [Fig. 1](#). The degraded performance of DS is attributable to its separate treatment of the real and imaginary parts, which is not equivalent to the block-coupled proximal mapping associated with $\sum_{j=1}^n \sqrt{x_j^2 + y_j^2}$. By contrast, the RI baseline remains an equivalent real-domain

Table 1
Average convergence time (s)
versus dimension N in Fig. 4.

N	CPNN (s)	RI (s)
500	0.10	0.11
1000	0.79	1.76
1500	1.96	4.41
2000	3.50	7.79
3000	8.04	18.89

reformulation, while CPNN keeps the same proximal structure directly in its native complex-domain form and avoids the additional $2n$ -dimensional representation.

We evaluate robustness using the NMSE (dB), $\text{NMSE} = 10 \log_{10} \left(\frac{\|z(t) - z_{\text{true}}\|_2^2}{\|z_{\text{true}}\|_2^2} \right)$. We consider three perturbation settings: measurement noise (SNR sweep), sensing-matrix mismatch (random row replacement), and signal perturbation ($z_{\text{true}} = z_{\text{sparse}} + \rho w$ with complex Gaussian w). As shown in Figs. 6(a)-8(a), CPNN exhibits robustness comparable to the RI baseline across all tested perturbation levels, and consistently outperforms DS, which suffers from a higher NMSE. The typical convergence curves in Figs. 6(b)-8(b) (SNR = 20 dB, mismatch rate = 10%, and $\rho = 0.02$) further show that CPNN achieves a faster decay in both the relative error and the objective gap than RI and DS under the same accuracy criterion.

Remark 5.1. In practice, choose L as a Lipschitz constant of f . For $f(z) = Mz - q$, one may take $L = \|M\|_2$. In Example 5.1 with $f(z) = A^H(Az - y)$, we use $L = \|A^H A\|_2 = \|A\|_2^2$, where $\|A\|_2$ can be estimated by $\text{svds}(A, 1)$ (or power iteration for large-scale problems). The constant θ is the modulus appearing in the $\theta - \Phi$ strong pseudomonotonicity condition and is used in the theory to state sufficient stability conditions such as $\eta L^2 < 2\theta$. As a stronger but easily checkable surrogate, when f is affine and globally strongly monotone, a practical estimate of the corresponding monotonicity modulus is $\lambda_{\min} = \left(\frac{M+M^H}{2} \right)$. For Example 5.1, however, the relevant property is interpreted in a restricted sparse-recovery sense rather than through a global full-space modulus; Since θ is not explicitly required by the implementation in Example 5.1, we recommend the scale-invariant choice $\eta = \frac{c}{L^2}$, $c \in (0, 1)$, and set $c = 0.95$ in our experiments. For the discrete-time implementation, we select the effective step size $h_n = \beta \psi_n \in (0, 1]$. In the noiseless experiments, we fix $\lambda = 0.1$, while we use an automatic rule for λ in the robustness experiment.

6. Conclusion

In this paper, we studied complex-valued mixed variational inequalities (CMVIs) with composite nonsmooth regularization and developed a complex-valued proximal neural network (CPNN) that operates intrinsically in $\mathbb{C}^n$. This design avoids passing through a $2n$ -dimensional real reformulation and enables the proximal mapping and the associated dynamics to be expressed directly in their native complex-domain form. On the theoretical side, we established the nonexpansiveness of the complex-valued proximal mapping under standard convexity assumptions, proved existence and uniqueness of the CMVIs solution, and derived a residual-based error bound under $\theta - \Phi$ strong pseudomonotonicity and Lipschitz continuity. Based on these results, we showed that the equilibrium points of the proposed CPNN coincide exactly with the solutions of the CMVIs, proved the well-posedness of the continuous-time dynamics, and established global exponential stability by combining the complex chain rule with Lyapunov analysis. We further analyzed a discrete-time implementation and obtained linear convergence. Numerical experiments in Section 5 validate the theoretical results and demonstrate that the proposed CPNN provides an effective and stable solver for CMVIs in $\mathbb{C}^n$. In future work, we will explore whether the current assumptions can be further relaxed and investigate whether similar

stability and convergence rate results can be established under weaker conditions.

CRedit authorship contribution statement

Yuanhua Wang: Writing – review & editing, Writing – original draft;
Siru Yin: Writing – review & editing; **Jun Li:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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